Lab_Syn_2.1

You have to synthesize ALU.v file using Design Compiler shell and generate Technology Dependent gate level netlists in Verilog format using standard cell libraries (typical, worst, best) inside std_cells folder:-

- scmetro_tsmc_cl013g_rvt_tt_1p2v_25c.db
- scmetro_tsmc_cl013g_rvt_ff_1p32v_m40c
- scmetro_tsmc_cl013g_rvt_ss_1p08v_125c

Steps: -

1. Copy Lab_Syn_2.1 folder from your computer into the virtual machine inside Labs directory
   - Open a terminal inside path /home/IC/Labs/Lab_Syn_2.1/syn and run syn_script.tcl script through design compiler shell using the following command in the terminal
     - dc_shell -f syn_script.tcl > syn.log
2. Every Time you synthesize a design, you need to check the synthesis log (syn.log) to be sure that: -
   - No Errors
   - No Warnings
   - No Latches
   - No Comb Loops

During Syntheize ALU.v using syn_script.tcl, you will find 4 issues, 2 of them will be solved by modifying in the syn_script .tcl and the other 2 issues needs modifications in ALU.v. Please fix the 4 issues and be sure that your synthesis log is free of Errors/Warnings, latches and loops.

Note: Can you find the rule of this command at the beginning of the script

define_design_lib work -path ./work